Milestone 1.3: Bayesian

INFO 521 · Machine Learning Foundations · project milestone brief · One Problem, Three Lenses · Lens 3 of 3

How to read this
This is the assignment brief for one project milestone. The Satisfactory criteria below are authoritative: every one must be met, and it is all-or-nothing rather than partial credit. A Not-Yet returns with targeted feedback and one revise cycle. Do the work in the notebook distributed through Classroom 50; this PDF is the brief, not the workspace.

Course-schedule placement. Part 1 of the project: due Week 5, the midpoint. This milestone covers M4 and is due Week 5. Gated by Checkpoint quiz 1.3 (conjugate posterior, Wk 4).

The third lens

You will now treat the weights as **uncertain quantities** with a prior, and update to a posterior given the data. Because a Gaussian prior on $\mathbf{w}$ is **conjugate** to the Gaussian-noise likelihood, the posterior is itself Gaussian and available in closed form, no sampling required (that comes in Part 2, once you break conjugacy).

Same ground rules: from-scratch NumPy/SciPy, no off-the-shelf solvers.

What “Satisfactory” means for 1.3

- ☐ You place a **Gaussian prior** on $\mathbf{w}$ and state the conjugate posterior (mean and covariance) in markdown.
- ☐ You implement the **posterior mean and covariance** in NumPy.
- ☐ You produce a **posterior-predictive** curve with **credible intervals**.
- ☐ **Play artifact**: a prior-strength sweep showing the posterior collapse toward the MLE as the prior weakens (or as data grows), with interpretation.
- ☐ **Checkpoint quiz 1.3 (identify the posterior for a conjugate pair)** passed.

Setup

```
import numpy as np
from info521 import data, plotting
plotting.use_course_style()
ds = data.load_clinical()
age = data.primary_predictor(ds)
y = ds["y"]
```

1 · Prior, posterior, and the closed form

In markdown, state your prior $\mathbf{w} \sim \mathcal{N}(\mathbf{0}, \tau^2\mathbf{I})$ and write the posterior $p(\mathbf{w} \mid \mathbf{X}, \mathbf{y}, \sigma^2)$. Label the posterior mean and covariance and say which term encodes the prior.

2 · Implement the posterior

```python
def posterior(X, y, sigma2, tau2):
    # TODO: return (mean, cov) of the Gaussian posterior over w.
    # No solver beyond linear algebra: this is exact.
    raise NotImplementedError
```

3 · Posterior predictive with credible intervals

For a grid of ages, propagate posterior uncertainty into predictions and shade a credible band with `plotting.credible_band(...)`.

```python
# TODO: predictive mean and variance at each grid age; shade the interval.
```

Play artifact: how strong is your prior?

Sweep the prior strength (e.g. τ^2 across several orders of magnitude) and, separately, the amount of data used (e.g. 10%, 50%, 100% of patients). Show:

1. As the prior **weakens**, the posterior mean approaches the MLE from 1.2.
2. As **data grows**, the posterior concentrates and the credible band narrows.

This is the coin-game intuition from lecture, reframed clinically: a strong prior resists the data; enough data overwhelms any prior.

Write 150–250 words interpreting the two sweeps.

```python
# TODO: the two sweeps + interpretation.
```

> **i Gate**
>
> Not Satisfactory until **Checkpoint quiz 1.3 (identify the posterior form for a given conjugate prior–likelihood pair)** is passed. The checkpoint is an open-book D2L quiz, due the week before this milestone; the auto-graded portion resolves on submission and written responses allow one revision.

Looking ahead

In **1.4 (synthesis)** you place all three lenses side by side. Then **Part 2** breaks the comfortable conjugacy you relied on here: binarize systolic BP into a hypertension label and the posterior is no longer closed-form, forcing approximate inference.